DSA 8070 R Session 9: Factor Analysis

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Intelligence Tests Example

Load the data summary

We use the built-in `ability.cov` dataset and convert its covariance matrix into a correlation matrix for factor analysis.

```
# Load the ability test data
data(ability.cov)

# Convert the covariance matrix to a correlation matrix
(cor <- cov2cor(ability.cov$cov))
```

```
##           general   picture    blocks      maze   reading     vocab
## general 1.0000000 0.4662649 0.5516632 0.3403250 0.5764799 0.5144058
## picture 0.4662649 1.0000000 0.5724364 0.1930992 0.2629229 0.2392766
## blocks  0.5516632 0.5724364 1.0000000 0.4450901 0.3540252 0.3564715
## maze    0.3403250 0.1930992 0.4450901 1.0000000 0.1839645 0.2188370
## reading 0.5764799 0.2629229 0.3540252 0.1839645 1.0000000 0.7913779
## vocab   0.5144058 0.2392766 0.3564715 0.2188370 0.7913779 1.0000000
```

```
factanal(x, factors, data = NULL, covmat = NULL, n.obs = NA,
         subset, na.action, start = NULL,
         scores = c("none", "regression", "Bartlett"),
         rotation = "varimax", control = NULL, ...)
```

Figure 1: factanal usage

Factor analysis

We will use the *factanal* command to perform factor analysis. The usage can be found below:

```
# Fit a one-factor model using maximum likelihood
(ability.FA <- factanal(factors = 1, covmat = ability.cov))
```

```
##
## Call:
## factanal(factors = 1, covmat = ability.cov)
##
## Uniquenesses:
## general picture  blocks    maze reading   vocab
##   0.535   0.853   0.748   0.910   0.232   0.280
##
## Loadings:
##         Factor1
## general 0.682
## picture 0.384
## blocks  0.502
## maze    0.300
## reading 0.877
## vocab   0.849
##
##                Factor1
## SS loadings      2.443
## Proportion Var   0.407
##
## Test of the hypothesis that 1 factor is sufficient.
## The chi square statistic is 75.18 on 9 degrees of freedom.
## The p-value is 1.46e-12
```

The goodness-of-fit test for the one-factor model yields a very small p-value (1.46×10^{-12}), providing strong evidence against the null hypothesis that one factor is sufficient. This suggests that a single factor does not adequately explain the covariance structure of the data.

```
# Update the fitted model to include two common factors
update(ability.FA, factors = 2)
```

```
##
## Call:
```

```
## factanal(factors = 2, covmat = ability.cov)
##
## Uniquenesses:
## general picture  blocks    maze reading   vocab
##   0.455   0.589   0.218   0.769   0.052   0.334
##
## Loadings:
##         Factor1 Factor2
## general 0.499   0.543
## picture 0.156   0.622
## blocks  0.206   0.860
## maze    0.109   0.468
## reading 0.956   0.182
## vocab   0.785   0.225
##
##                Factor1 Factor2
## SS loadings      1.858   1.724
## Proportion Var   0.310   0.287
## Cumulative Var   0.310   0.597
##
## Test of the hypothesis that 2 factors are sufficient.
## The chi square statistic is 6.11 on 4 degrees of freedom.
## The p-value is 0.191
```

Stock Price Example

Load the stock price data

We load the weekly stock returns for five companies, assign descriptive variable names, and inspect the first few observations.

```r
# Read the stock returns data
url <- "https://pages.stat.wisc.edu/~rich/JWMULT06dat/T8-4.DAT"
stock <- read.table(url, sep = "\t", header = FALSE)

# Assign company names
colnames(stock) <- c("JP Morgan", "Citibank", "Wells Fargo",
                     "Royal Dutch", "Exxon")

# Display the first few observations
head(stock)
```

```
##    JP Morgan   Citibank Wells Fargo Royal Dutch      Exxon
## 1  0.0130338 -0.0078431  -0.0031889  -0.0447693  0.0052151
## 2  0.0084862  0.0166886  -0.0062100   0.0119560  0.0134890
## 3 -0.0179153 -0.0086393   0.0100360   0.0000000 -0.0061428
## 4  0.0215589 -0.0034858   0.0174353  -0.0285917 -0.0069534
## 5  0.0108225  0.0037167  -0.0101345   0.0291900  0.0409751
## 6  0.0101713 -0.0121978  -0.0083768   0.0137083  0.0029895
```

Summary statistics

```
(Xbar <- colMeans(stock))
```

```
##    JP Morgan     Citibank  Wells Fargo  Royal Dutch        Exxon
## 0.0010627806 0.0006554204 0.0016260816 0.0040491252 0.0040386417
```

```
(S <- cov(stock))
```

```
##                JP Morgan     Citibank  Wells Fargo  Royal Dutch        Exxon
## JP Morgan   4.332695e-04 0.0002756679 1.590265e-04 6.411929e-05 8.896616e-05
## Citibank    2.756679e-04 0.0004387172 1.799737e-04 1.814512e-04 1.232623e-04
## Wells Fargo 1.590265e-04 0.0001799737 2.239722e-04 7.341348e-05 6.054612e-05
## Royal Dutch 6.411929e-05 0.0001814512 7.341348e-05 7.224964e-04 5.082772e-04
## Exxon       8.896616e-05 0.0001232623 6.054612e-05 5.082772e-04 7.656742e-04
```

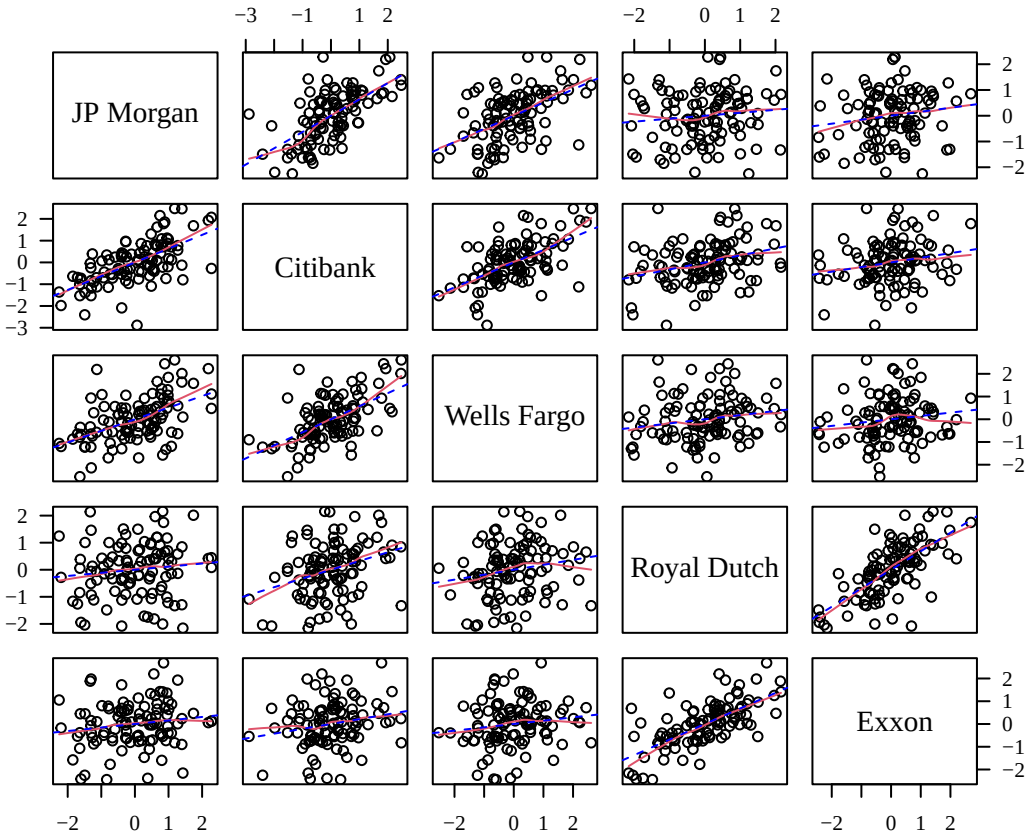
Scatter plot matrix

We standardize the stock returns to have mean zero and unit variance, then use a scatterplot matrix to explore pairwise relationships among the five companies.

```
# Standardize the stock returns
stock_std <- scale(stock, center = TRUE, scale = TRUE)
stock_std <- as.data.frame(stock_std)

# Set graphical parameters
par(las = 1, mgp = c(2.4, 1, 0), mar = c(3.5, 3.5, 1, 0.6),
    family = "serif")

# Scatterplot matrix with smooth trends and fitted regression lines
pairs(stock_std, labels = names(stock),
      panel = function(x, y) {
        panel.smooth(x, y)
        abline(lsfit(x, y), lty = 2, col = "blue")
      })
```



Compute PCs from the correlation matrix

We perform PCA on the standardized stock returns and examine the proportion of variance explained by each principal component. We then visualize the individual and cumulative proportions to help determine how many components to retain.

```
# Compute the sample correlation matrix
(s.cor <- var(stock_std))
```

```
##             JP Morgan  Citibank Wells Fargo Royal Dutch     Exxon
## JP Morgan   1.0000000 0.6322878   0.5104973   0.1146019 0.1544628
## Citibank    0.6322878 1.0000000   0.5741424   0.3222921 0.2126747
## Wells Fargo 0.5104973 0.5741424   1.0000000   0.1824992 0.1462067
## Royal Dutch 0.1146019 0.3222921   0.1824992   1.0000000 0.6833777
## Exxon       0.1544628 0.2126747   0.1462067   0.6833777 1.0000000
```

```
# Perform PCA using standardized stock returns
s.pca <- prcomp(stock, scale = TRUE, center = TRUE)

# Examine the principal component loadings
s.pca$rotation
```

```
##                   PC1        PC2         PC3        PC4         PC5
## JP Morgan   -0.4690832 -0.3680070 -0.60431522 -0.3630228  0.38412160
## Citibank    -0.5324055 -0.2364624 -0.13610618  0.6292079 -0.49618794
```

```
## Wells Fargo -0.4651633 -0.3151795  0.77182810 -0.2889658  0.07116948
## Royal Dutch -0.3873459  0.5850373  0.09336192  0.3812515  0.59466408
## Exxon       -0.3606821  0.6058463 -0.10882629 -0.4934145 -0.49755167
```

```r
# Compute the variance explained by each component
s <- var(s.pca$x)
(Proportion.std <- round(diag(s) / sum(diag(s)), 3))
```

```
##   PC1   PC2   PC3   PC4   PC5
## 0.487 0.281 0.100 0.080 0.051
```

```r
(Cumulative.std <- round(cumsum(diag(s)) / sum(diag(s)), 3))
```

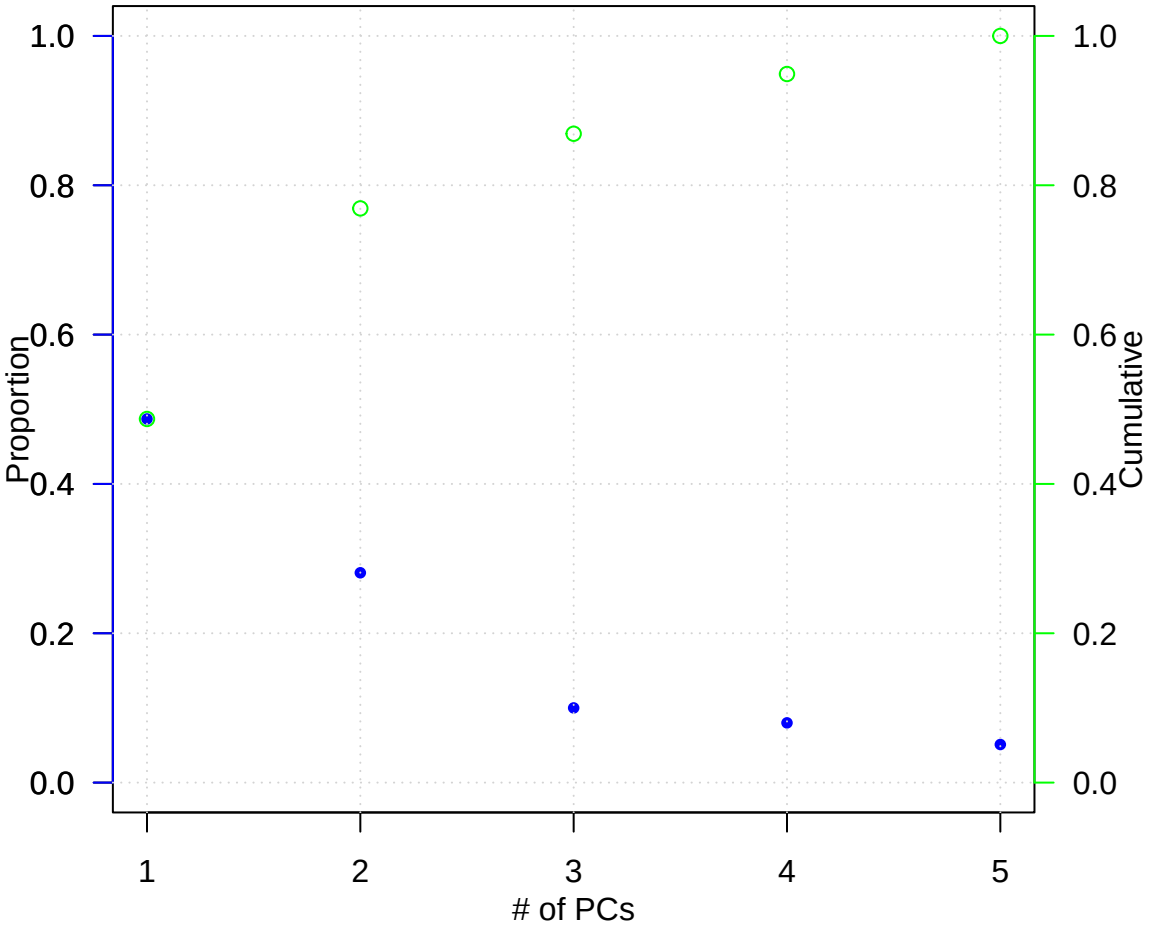
```
##   PC1   PC2   PC3   PC4   PC5
## 0.487 0.769 0.869 0.949 1.000
```

```r
# Plot individual and cumulative proportions of explained variance
p <- 5
par(las = 1, mgp = c(2, 1, 0), mar = c(3, 3, 1, 3))

plot(1:p, Proportion.std, xlab = "# of PCs", ylim = c(0, 1),
     ylab = "Proportion", pch = 16, cex = 0.8,
     xaxt = "n", col = "blue")

axis(1, at = 1:p)
mtext("Cumulative", 4, las = 0, line = 2)
axis(4, col = "green")
axis(2, col = "blue")
grid()

# Add cumulative explained variance
points(1:p, Cumulative.std, cex = 1, col = "green")
```



Factor Loadings and specific variances

We fit a two-factor model using the principal component method. We estimate the factor loadings and specific variances, then examine the residual correlation matrix to assess how well the model reproduces the observed correlations.

```
# Extract eigenvalues and eigenvectors from PCA
lambda <- s.pca$sdev^2
e <- s.pca$rotation

# Estimate factor loadings for m = 2
sqrt(lambda[1]) * e[, 1]
```

```
##   JP Morgan    Citibank Wells Fargo Royal Dutch       Exxon
##  -0.7323218  -0.8311791  -0.7262022  -0.6047155  -0.5630885
```

```
sqrt(lambda[2]) * e[, 2]
```

```
##   JP Morgan    Citibank Wells Fargo Royal Dutch       Exxon
##  -0.4365209  -0.2804859  -0.3738582   0.6939569   0.7186401
```

```r
# Estimate specific variances
sVar <- diag(s.cor - (lambda[1] * e[, 1] %*% t(e[, 1]) +
                      lambda[2] * e[, 2] %*% t(e[, 2])))

# Examine the residual correlation matrix
round(s.cor - (lambda[1] * e[, 1] %*% t(e[, 1]) +
               lambda[2] * e[, 2] %*% t(e[, 2]) +
               diag(sVar)), 2)
```

```
##             JP Morgan Citibank Wells Fargo Royal Dutch Exxon
## JP Morgan        0.00    -0.10       -0.18       -0.03  0.06
## Citibank        -0.10     0.00       -0.13        0.01 -0.05
## Wells Fargo     -0.18    -0.13        0.00        0.00  0.01
## Royal Dutch     -0.03     0.01        0.00        0.00 -0.16
## Exxon            0.06    -0.05        0.01       -0.16  0.00
```

Interpretation: Ideally, the off-diagonal residual correlations should be close to zero, indicating that the two common factors adequately capture the dependence among the five stocks.

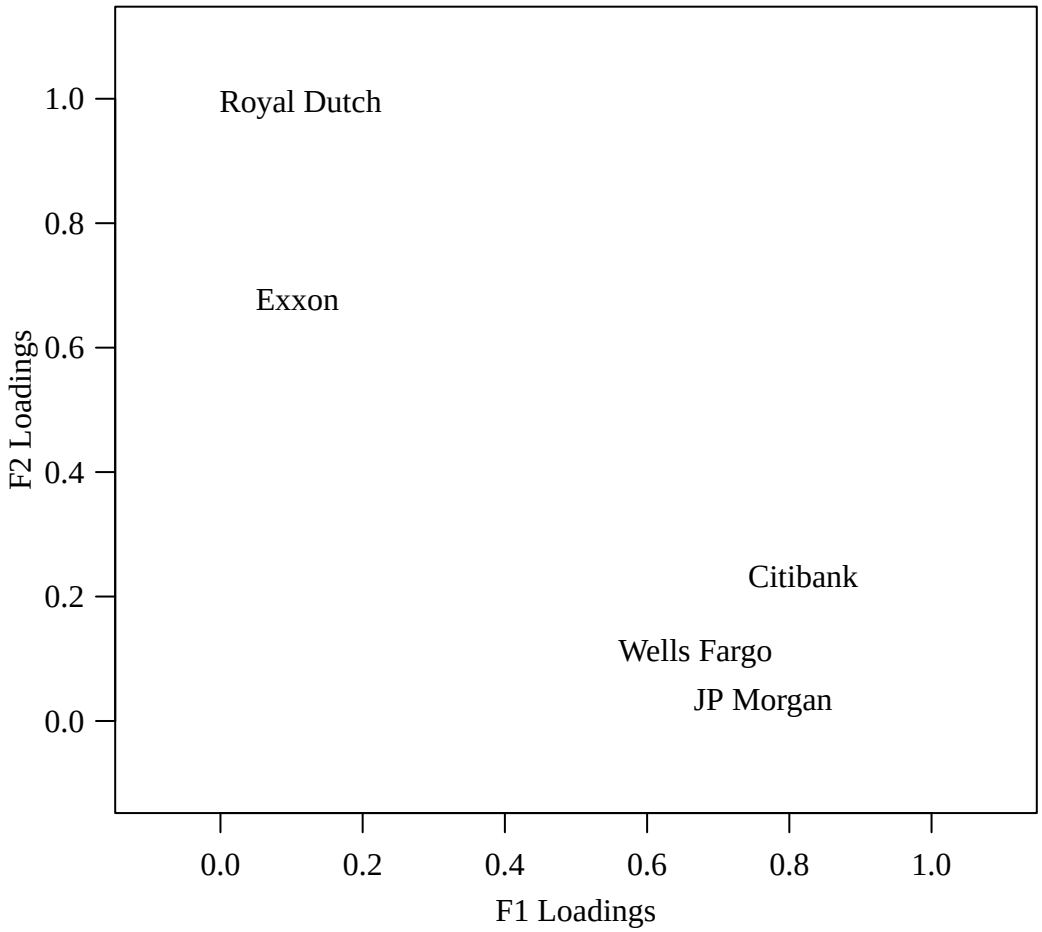
MLE

We fit a two-factor model using maximum likelihood estimation with `varimax` rotation (the default in *factanal()*). We visualize the factor loadings to explore how the five stocks relate to the two factors, then examine the residual correlation matrix to assess model fit.

```r
# Fit a two-factor model using maximum likelihood
(stock.fac <- factanal(stock, factors = 2, rotation = "varimax"))
```

```
##
## Call:
## factanal(x = stock, factors = 2, rotation = "varimax")
##
## Uniquenesses:
##   JP Morgan    Citibank Wells Fargo Royal Dutch       Exxon
##       0.417       0.275       0.542       0.005       0.530
##
## Loadings:
##             Factor1 Factor2
## JP Morgan   0.763
## Citibank    0.819   0.232
## Wells Fargo 0.668   0.108
## Royal Dutch 0.113   0.991
## Exxon       0.108   0.677
##
##                Factor1 Factor2
## SS loadings      1.725   1.507
## Proportion Var   0.345   0.301
## Cumulative Var   0.345   0.646
##
## Test of the hypothesis that 2 factors are sufficient.
## The chi square statistic is 1.97 on 1 degree of freedom.
## The p-value is 0.16
```

```
# Plot the estimated factor loadings
par(las = 1, mgp = c(2, 0.8, 0), mar = c(3, 3, 1, 3),
    family = "serif")
plot(stock.fac$loadings, xlab = "F1 Loadings", ylab = "F2 Loadings",
     type = "n", xlim = c(-0.1, 1.1), ylim = c(-0.1, 1.1))
text(stock.fac$loadings, labels = colnames(stock))
```




```
# Compute the model-implied correlation matrix
pred <- (stock.fac$loadings %*% t(stock.fac$loadings)) +
        diag(stock.fac$uniquenesses)

# Examine the residual correlation matrix
(resid <- s.cor - pred)
```

```
##                  JP Morgan      Citibank   Wells Fargo   Royal Dutch
## JP Morgan     1.055860e-07  7.496780e-06 -2.564223e-03 -3.325561e-04
## Citibank      7.496780e-06  3.255673e-08  1.608871e-03  2.116218e-04
## Wells Fargo  -2.564223e-03  1.608871e-03  5.157373e-08 -9.518792e-06
## Royal Dutch  -3.325561e-04  2.116218e-04 -9.518792e-06 -1.559500e-06
## Exxon         5.198222e-02 -3.307885e-02  5.547153e-04  1.218853e-04
##                        Exxon
## JP Morgan     5.198222e-02
## Citibank     -3.307885e-02
```

```
## Wells Fargo  5.547153e-04
## Royal Dutch  1.218853e-04
## Exxon        2.670491e-07
```